

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2015

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 12 March 2016

Time: 13.00-15.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books are NOT allowed.
4. A calculator is allowed.
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 11 pages (including this page), 3 parts (59 questions) and a total score of 75 points.

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Part 1 (25 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Capacitance is directly proportional to the distance between plates.
2. When charging capacitor, the voltage of capacitor increases exponentially.
3. The steady-state level of the capacitor current can be found by substituting the short-circuit equivalent.
4. In RC charging network, the capacitor will be fully charged after 5τ .
5. The unit of τ is in second.
6. When charging inductor, the voltage of inductor increases exponentially.
7. When closing switch of RL DC circuit, electric current can flow immediately at the beginning.
8. The time constant for an inductor circuit equals its inductance divided by its resistance.
9. An ideal inductor acts like an open circuit to dc current.
10. The current across a capacitor leads the voltage through it by 90° .
11. Inductive reactance increases inversely to frequency.
12. Power factor of resistive circuit is equal to one.
13. For a purely capacitive element, the voltage and the current through the element are in phase.
14. A capacitive network has a leading power factor.
15. The magnitude of polar form is in effective value.
16. Resistor value is independent from frequency.
17. At very high frequency, inductor acts as an open circuit.

18. The unit of capacitive reactance is in Ohm.
19. The unit of average power is in VAR.
20. The unit of power factor is in degree.
21. The voltage divider rule for dc circuits can be used for ac circuits.
22. For a parallel ac circuit with reactive elements, the current in coils or capacitors are always in the same direction on a phasor diagram.
23. In RC series circuit, electric current lags the voltage at power supply by 90 degrees.
24. In series network, total admittance is the sum of the individual admittance.
25. The total impedance has a positive phase angle is a clear indication that the network is capacitive in nature.

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Part 2 (30 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Find the value of capacitor of a small capacitor below.

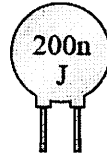


FIGURE 1

- (a) 200 nF $\pm$ 5% (b) 200 nF $\pm$ 10% (c) 200 pF $\pm$ 5% (d) 200 pF $\pm$ 10%

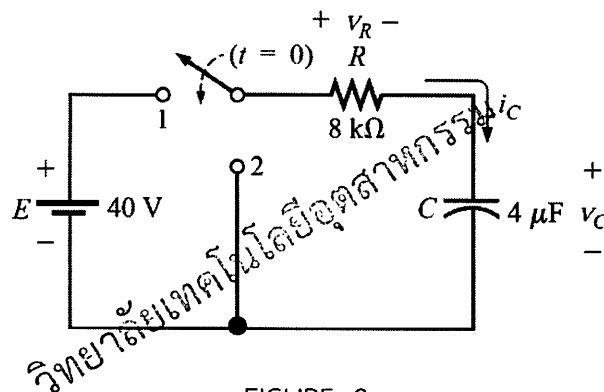


FIGURE 2

2. What is the time constant of the circuit from figure 2?
 (a) 32 ms (b) 2 ps (c) 0.5 ns (d) 4 μ s
3. From figure 2, what is the capacitor voltage at charging state?
 (a) $40e^{t/\tau}$ [V] (b) $40e^{-t/\tau}$ [V] (c) $40(1 - e^{t/\tau})$ [V] (d) $40(1 - e^{-t/\tau})$ [V]
4. From figure 2, what is the capacitor current at charging state?
 (a) $5e^{t/\tau}$ [mA] (b) $5e^{-t/\tau}$ [mA] (c) $10e^{t/\tau}$ [mA] (d) $10e^{-t/\tau}$ [mA]
5. From figure 2, what is the capacitor voltage at discharging state?
 (a) $40e^{t/\tau}$ [V] (b) $40e^{-t/\tau}$ [V] (c) $40(1 - e^{t/\tau})$ [V] (d) $40(1 - e^{-t/\tau})$ [V]
6. From figure 2, what is the capacitor current at discharging state?
 (a) $5e^{t/\tau}$ [mA] (b) $5e^{-t/\tau}$ [mA] (c) $-5e^{t/\tau}$ [mA] (d) $-5e^{-t/\tau}$ [mA]

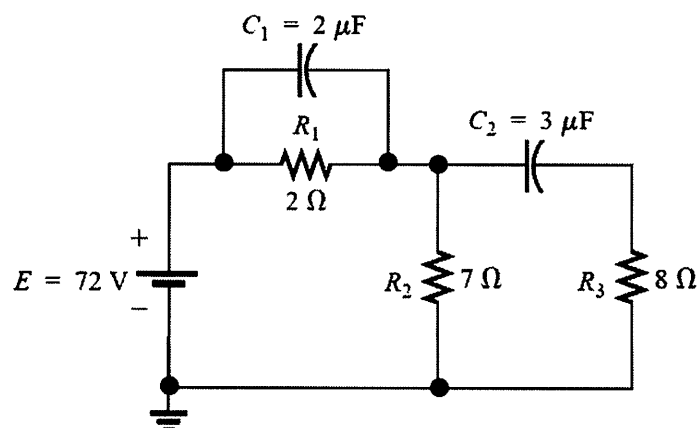


FIGURE 3

7. From figure 3, find the voltage across the capacitor C_1 .
 (a) 56 V (b) 16 V (c) 46.9 V (d) 25.1 V
8. From figure 3, find the voltage across the capacitor C_2 .
 (a) 56 V (b) 16 V (c) 46.9 V (d) 25.1 V
9. From figure 3, find the energy store in capacitor C_1 .
 (a) 3136 μJ (b) 256 μJ (c) 2199.6 μJ (d) 630 μJ
10. From figure 3, find the energy store in capacitor C_2 .
 (a) 4704 μJ (b) 384 μJ (c) 2199.6 μJ (d) 945 μJ
11. Find the capacitance of a parallel plate capacitor if the area of each plate is 0.075 m^2 and the distance between the plates is 1.77 mm. The dielectric is air ($\epsilon_r = 1.0006$) where $\epsilon_0 = 8.85 \times 10^{-12}\text{ F/m}$.
 (a) 0.2089 pF (b) 208.98 pF (c) 0.375 pF (d) 375.225 pF
12. What is the type of the following inductor?



FIGURE 4

- (a) variable (b) air-core (c) iron-core (d) ceramic

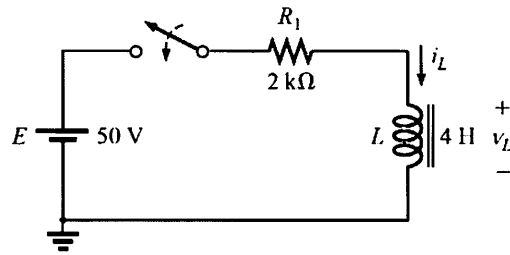


FIGURE 5

13. What is the time constant of the circuit from figure 5?

- (a) 2 ms (b) 0.5 ms (c) 0.5 ks (d) 8 ks

14. From figure 5, what is the inductor current?

- (a) $25e^{v/\tau}$ [mA] (b) $25e^{-v/\tau}$ [A] (c) $25(1 - e^{-v/\tau})$ [mA] (d) $25(1 - e^{v/\tau})$ [A]

15. From figure 5, what is the inductor voltage?

- (a) $50e^{v/\tau}$ [V] (b) $50e^{-v/\tau}$ [V] (c) $50(1 - e^{-v/\tau})$ [V] (d) $50(1 - e^{v/\tau})$ [V]

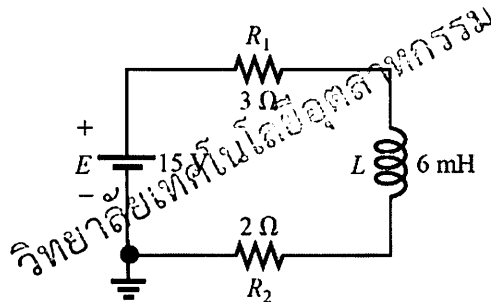


FIGURE 6

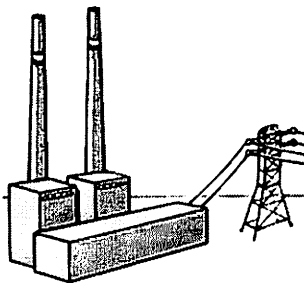
16. From figure 6, find the inductor current.

- (a) 0 A (b) 3 A (c) 5 A (d) 7.5 A

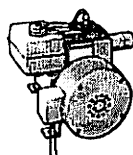
17. From figure 6, find the energy store in inductor.

- (a) 0 mJ (b) 27 mJ (c) 75 mJ (d) 168.75 mJ

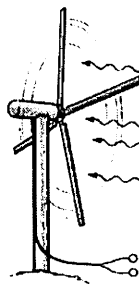
18. Which one of following power generations has the output power different from others?



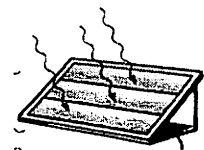
(a)



(b)



(c)



(d)

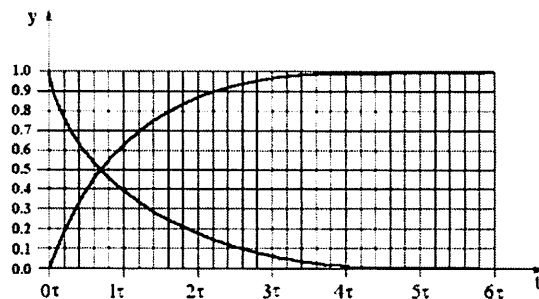


FIGURE 7

19. From figure 7, which of the following equations is increasing curve?
 (a) $y = e^{t/\tau}$ (b) $y = e^{-t/\tau}$ (c) $y = (1 - e^{-t/\tau})$ (d) $y = (1 - e^{t/\tau})$
20. From figure 7, which of the following equations is decreasing curve?
 (a) $y = e^{t/\tau}$ (b) $y = e^{-t/\tau}$ (c) $y = (1 - e^{-t/\tau})$ (d) $y = (1 - e^{t/\tau})$
21. Find the capacitance of a parallel-plate capacitor if a 35 V potential difference exists with 2,000 μC of charge.
 (a) 42 μF (b) 73 μF (c) 57 μF (d) 28 μF
22. Find the capacitance of a parallel-plate air dielectric capacitor if the area of each of its plates is 0.1 m^2 and the distance between the plates is 2.0 mm.
 (a) 1.44 nF (b) 0.62 nF (c) 0.44 nF (d) 0.14 nF
23. What is the inductive reactance at 800 Hz of a 1 mH inductor?
 (a) 12 Ω (b) 5.0 Ω (c) 20 Ω (d) 0.2 Ω
24. At what frequency does a 10 μF capacitor have a reactance of 100 Ω ?
 (a) 15.9 kHz (b) 1.59 kHz (c) 159 Hz (d) 1.59 MHz
25. The voltage across a 100 mH coil is $v = 100 \sin 50t$. Which of these expressions describes the current?
 (a) $2000 \sin(50t - 90^\circ)$ (b) $20 \sin(50t + 90^\circ)$
 (c) $20 \sin 50t$ (d) $20 \sin(50t - 90^\circ)$
26. Which one of the following values is equivalent to $1 \angle 180^\circ$?
 (a) $-1 \angle 0^\circ$ (b) $-1 \angle 90^\circ$ (c) $1 \angle 360^\circ$ (d) $1 \angle -90^\circ$
27. What is the effective (rms) value of $v = 23 \sin 400t$?
 (a) 400 V (b) 23 V (c) 16.26 V (d) 11.5 V
28. Which one of the following phasor domain expressions is equivalent to the time domain expression $50 \cos(\omega t - 15^\circ)$?
 (a) $35.35 \angle -15^\circ$ (b) $35.35 \angle 15^\circ$
 (c) $35.35 \angle 105^\circ$ (d) $35.35 \angle -105^\circ$

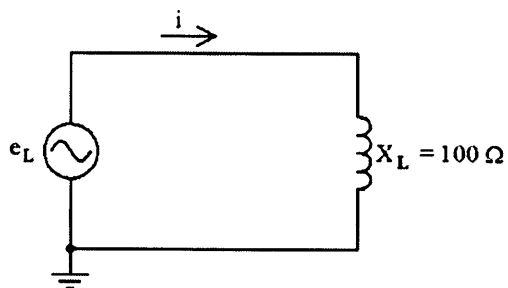


Figure 8

29. See Figure 8. What is the coil current if e_L is $500 \sin(50t + 20^\circ)$?

- (a) $5 \sin(50t + 20^\circ)$ (b) $5 \sin(50t - 70^\circ)$
 (c) $5 \sin(50t + 110^\circ)$ (d) $5 \sin 50t$

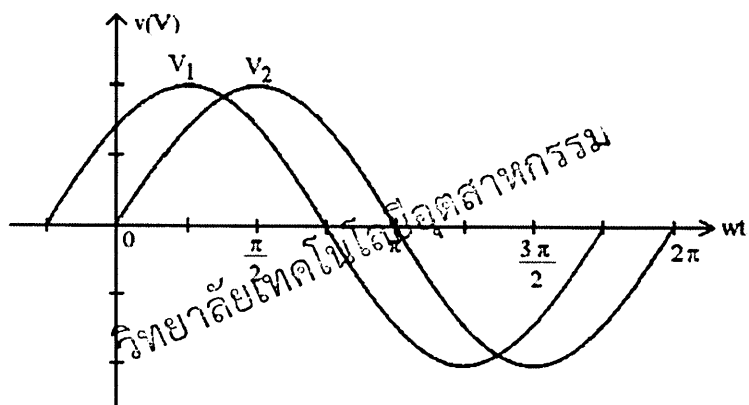


Figure 9

30. See Figure 9, what relationship exists between voltages v_1 and v_2 ?

- (a) v_2 leads v_1 by $\pi/4$. (b) v_2 lags v_1 by 90° .
 (c) v_1 leads v_2 by 45° . (d) v_1 lags v_2 by $\pi/4$.

Part 3 (20 points) Calculation

Instruction: Shows the mathematical expression and answers of following problems.

1. Find the total ADMITTANCE (Y_T) and IMPEDANCE (Z_T) of the circuit of Figure 10.

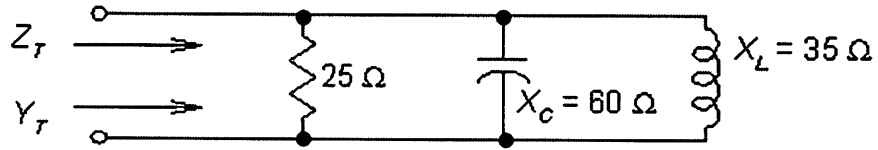


Figure 10

2. Evaluate the expression $(4 - j5) / (39 \angle 42^\circ + 4 + j5)$ and express the answer in polar and rectangular forms. Show all of steps not only the final answer.

3. Calculate the currents I_1 and I_2 of Figure 11 in phasor form using the current divider rule.

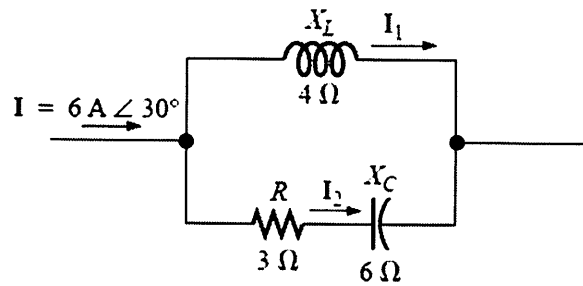


Figure 11

4. From the circuit in Figure 12, find V_{ab} .

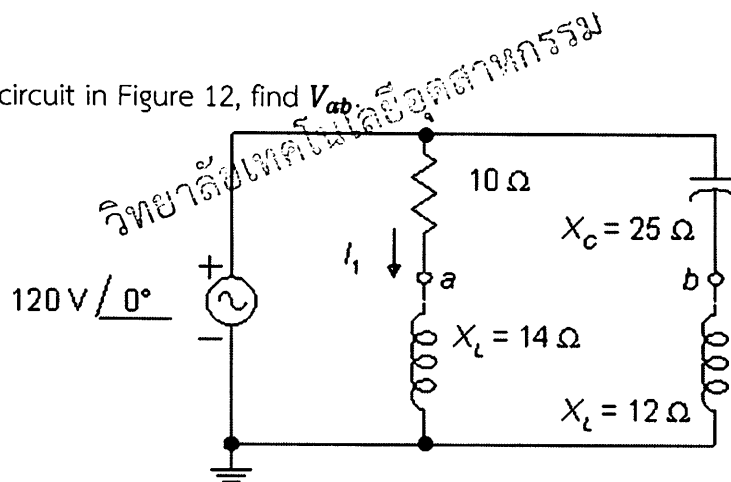


Figure 12

Answer Sheet

Name: _____ ID: _____

Subject: 341152 Electric Circuits II

Part 1

	True	False
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3.		
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Part 2

	a	b	c	d
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